Grade 11 – Term 1
C8 Practice Activity with Solutions

Contents

1 We use Student's t-distribution to model standardized sample means. 1

2 Use Student's t-distribution tables to find probabilities and critical values. 4

This activity assesses the following criterion.

C8. Students use Student's t-distribution tables, degrees of freedom, and areas in one or both tails to determine critical values, central probabilities, and tail probabilities or probability ranges.

Open the normal and Student's t reference tables

Use the Student's t-distribution sheet on the second page of the linked reference document throughout this activity.

1 We use Student's t-distribution to model standardized sample means.

Problems:

1.1. The shape depends on the degrees of freedom.

1.2. Symmetry connects tail areas and central areas.

1.3. Read the linked Student's t-distribution table.

1.4. Study three worked table examples.

1.1 The shape depends on the degrees of freedom.

Assume $n \geq 2$ independent random observations from a normal population. The population mean μ and standard deviation $\sigma > 0$ are fixed parameters; $\bar{X}$ and S are the random sample mean and sample standard deviation. When unknown σ is estimated by S,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_\nu, \qquad \nu = n - 1,$$

where n is sample size and ν (df) is degrees of freedom. The denominator estimates the standard error of $\bar{X}$; T has no units. t_ν denotes Student's t-distribution, continuous and symmetric about zero with total probability 1. Its tails are heavier than those of $\mathcal{N}(0, 1)$, the standard normal distribution (mean 0, standard deviation 1). As ν increases, t approaches the standard normal distribution and its positive critical values decrease toward normal values.

1.2 Symmetry connects tail areas and central areas.

For $T \sim t_\nu$, α is an upper-tail probability and $t^*_{\alpha,\nu}$ its positive critical value:

$$P(T > t^*_{\alpha,\nu}) = \alpha, \qquad 0 < \alpha < \tfrac{1}{2}.$$

Write $P(\cdot)$ for probability and $\sim$ for follows the distribution. A cutoff c is a fixed boundary (possibly negative); $\pm c$ denotes both $-c$ and c, and $|Y|$ is the absolute value of Y. For continuous distributions, endpoints do not affect probabilities: $<$ and $\leq$ (or $>$ and $\geq$) are interchangeable. $\approx$ means approximate equality.

For $Y = T$ symmetric about zero, $c > 0$, and $\alpha = P(Y > c)$:

$$P(Y < -c) = \alpha, \qquad P(Y < c) = 1 - \alpha, \quad P(Y > -c) = 1 - \alpha,$$
$$P(|Y| > c) = 2\alpha, \quad P(-c < Y < c) = 1 - 2\alpha.$$

A combined two-tail area q uses $\alpha = q/2$; a central area p uses $\alpha = (1-p)/2$, where p, q are probabilities. The symmetric cutoffs are $\pm c$. For any real endpoints $a < b$,

$$P(a < Y < b) = 1 - P(Y \leq a) - P(Y \geq b), \qquad P(Y > c) = 1 - P(Y \leq c).$$

1.3 Read the linked Student's t-distribution table.

On reference page 2, select row ν and column α: the entry is $t^*_{\alpha,\nu}$, a cutoff, not a probability. Columns are $0.30, 0.20, 0.10, 0.05, 0.025, 0.01, 0.005$; all df needed here have rows. Convert the requested area to one upper tail first; negate the entry for a lower-tail cutoff.

Entries are rounded to three decimals: report cutoffs to three decimals and probabilities at printed cutoffs as approximations. For adjacent upper-tail columns with $\alpha_1 > \alpha_2$,

$$t^*_{\alpha_1,\nu} < c < t^*_{\alpha_2,\nu} \quad \Longrightarrow \quad \alpha_2 < P(T > c) < \alpha_1.$$

These are probability bounds, not an interpolated estimate. Symmetry, doubling, and complements give other bounds; for example, $1 - 2\alpha_1 < P(-c < T < c) < 1 - 2\alpha_2$. Allow for entry rounding if a cutoff is extremely close to a printed value; the between-entry cutoffs here are well separated from that uncertainty.

1.4 Study three worked table examples.

C8 Example 1: Find a positive cutoff and a negative cutoff.

An analyst models a standardized mean return from 10 independent observations. Find the positive cutoff exceeded by the standardized sample mean with probability 0.05. Also find the negative cutoff below which the standardized sample mean falls with probability 0.05.

C8 Worked-example solution

Here $\nu = 10 - 1 = 9$. Each request concerns one tail. Row 9, column $\alpha = 0.05$, gives $t^*_{0.05,9} = 1.833$ to three decimal places. Thus the positive cutoff is approximately 1.833 and the negative cutoff is approximately -1.833. Symmetry gives

$$P(T < -1.833) = P(T > 1.833) \approx 0.05.$$

C8 Example 2: Find symmetric cutoffs for a central probability.

A standardized household-expenditure mean uses 15 observations and has 14 degrees of freedom. Find symmetric cutoffs enclosing a central probability of 0.95.

C8 Worked-example solution

$T \sim t_\nu$ with $\nu = n - 1 = 15 - 1 = 14$. The outside probability is $1 - 0.95 = 0.05$, shared equally between the two tails. Use $\alpha = 0.05/2 = 0.025$. Row 14 gives

$$t^*_{0.025,14} = 2.145, \qquad P(-2.145 < T < 2.145) \approx 1 - 2(0.025) = 0.95.$$

The cutoffs are approximately -2.145 and 2.145; equivalently, $P(|T| > 2.145) \approx 0.05$.

C8 Example 3: Bound probabilities using symmetry.

A standardized delivery-time mean uses 20 observations and has 19 degrees of freedom. Give probability bounds for the standardized sample mean exceeding 2.30, falling below negative 2.30, and falling outside the symmetric cutoffs negative 2.30 and 2.30.

C8 Worked-example solution

$T \sim t_\nu$ with $\nu = n - 1 = 20 - 1 = 19$. In row 19,

$$t^*_{0.025,19} = 2.093 < 2.30 < 2.539 = t^*_{0.01,19}.$$

Larger positive cutoffs leave smaller upper-tail areas. Symmetry then gives

$$0.01 < P(T > 2.30) < 0.025, \qquad 0.01 < P(T < -2.30) < 0.025.$$

Doubling the equal tail areas and taking the complement gives

$$0.02 < P(|T| > 2.30) < 0.05, \qquad 0.95 < P(-2.30 < T < 2.30) < 0.98.$$



2 Use Student's t-distribution tables to find probabilities and critical values.

For each problem, record the degrees of freedom, classify the requested area, and identify the upper-tail column or the two columns bounding it. All sample-mean models below assume independent random observations from a normal population, with unknown population standard deviation replaced by the sample standard deviation. Use only the Student's t-distribution table on the second page of the linked reference sheet. Solutions follow each problem so that you can check your work after an independent attempt.

Problems:

2.1. Find a right-tail cutoff for investment returns.

2.2. Find a left-tail cutoff for household expenditures.

2.3. Find two-tail cutoffs for delivery times.

2.4. Find central cutoffs for production costs.

2.5. Read a printed entry for standardized mean wages.

2.6. Bound a right-tail probability for sales.

2.7. Bound two-tail and central probabilities for business revenues.

2.8. Interpret a negative standardized mean price.

2.9. Interpret verbal probabilities for small business costs.

2.10. Determine the sample size for payroll records.

2.11. Find probability ranges for revenue planning.

2.1 Find a right-tail cutoff for investment returns.

A financial analyst uses 10 daily investment returns. The standardized sample mean follows Student's t-distribution with 9 degrees of freedom.

Questions/Tasks

C8 Find the positive cutoff that the standardized sample mean exceeds with probability 0.10.

C8 Solution

$T \sim t_\nu$, where $\nu = n - 1 = 10 - 1 = 9$. This is one right tail. Use row 9 and column $\alpha = 0.10$: $t^*_{0.10,9} = 1.383$, so the positive cutoff is approximately 1.383.

2.2 Find a left-tail cutoff for household expenditures.

A survey uses 12 household expenditures. The standardized sample mean follows Student's t-distribution with 11 degrees of freedom.

Questions/Tasks

C8 Find the negative cutoff below which the standardized sample mean falls with probability 0.025.

C8 Solution

$T \sim t_\nu$, where $\nu = n - 1 = 12 - 1 = 11$. By symmetry, use the upper-tail column $\alpha = 0.025$ in row 11, then change the sign: $t^*_{0.025,11} = 2.201$, so $c \approx -2.201$.



2.3 Find two-tail cutoffs for delivery times.

A logistics company uses delivery times from 15 deliveries to model a standardized sample mean.

Questions/Tasks



C8 Determine the degrees of freedom and find the positive cutoff whose matching symmetric cutoffs leave a combined probability of 0.10 outside them. State both cutoffs for the standardized sample mean and the probability in each tail.

C8 Solution

$T \sim t_\nu$, where $\nu = n - 1 = 15 - 1 = 14$. The combined two-tail area is 0.10, so each tail has $\alpha = 0.05$. Row 14 gives $t^*_{0.05,14} = 1.761$; the cutoffs are approximately -1.761 and 1.761.



2.4 Find central cutoffs for production costs.

A manufacturer uses production costs from 20 independent batches to model a standardized sample mean.

Questions/Tasks



C8 Find symmetric negative and positive cutoffs for the standardized sample mean enclosing a central probability of 0.98. State the combined probability outside them.

C8 Solution

$T \sim t_\nu$, where $\nu = n - 1 = 20 - 1 = 19$. The outside area is 0.02, so $\alpha = 0.01$. Row 19 gives $t^*_{0.01,19} = 2.539$, so the cutoffs are approximately -2.539 and 2.539.



2.5 Read a printed entry for standardized mean wages.

A business models a standardized mean wage using a sample of 25 employees.

Questions/Tasks



C8 Use the table to estimate the probability that the standardized sample mean exceeds 2.064 and the probability that it is below 2.064.

C8 Solution

$T \sim t_\nu$, where $\nu = n - 1 = 25 - 1 = 24$. The entry $t^*_{0.025,24} = 2.064$ is in column $\alpha = 0.025$. Taking the complement gives the probability below the cutoff. Therefore $P(T > 2.064) \approx 0.025$ and $P(T < 2.064) \approx 0.975$.



2.6 Bound a right-tail probability for sales.

A retailer uses sales from 30 randomly selected trading days to model a standardized sample mean.

Questions/Tasks



C8 Give lower and upper bounds for the probability that the standardized sample mean exceeds 1.90. Identify the two table entries used.

C8 Solution

$T \sim t_\nu$, where $\nu = n - 1 = 30 - 1 = 29$. Since $1.699 < 1.90 < 2.045$, the bounding entries are $t^*_{0.05,29} = 1.699$ and $t^*_{0.025,29} = 2.045$. Larger cutoffs leave smaller upper-tail probabilities. Hence $0.025 < P(T > 1.90) < 0.05$.



2.7 Bound two-tail and central probabilities for business revenues.

A revenue model uses standardized sample means from 12 independent businesses.

Questions/Tasks



C8 Give lower and upper bounds for the combined probability that the standardized sample mean falls below negative 2.50 or above 2.50. Also give probability bounds for the standardized sample mean lying between negative 2.50 and 2.50.

C8 Solution

$T \sim t_\nu$, where $\nu = n - 1 = 12 - 1 = 11$. Since $2.201 < 2.50 < 2.718$, $0.01 < P(T > 2.50) < 0.025$. Doubling gives $0.02 < P(|T| > 2.50) < 0.05$. The central area therefore satisfies $0.95 < P(-2.50 < T < 2.50) < 0.98$.



2.8 Interpret a negative standardized mean price.

A price study models standardized mean prices from samples of 15 shops.

Questions/Tasks



C8 Give probability bounds for the standardized sample mean falling below negative 2.30 and for it exceeding negative 2.30. Explain the role of symmetry.

C8 Solution

$T \sim t_\nu$, where $\nu = n - 1 = 15 - 1 = 14$. The entries $2.145 < 2.30 < 2.624$ imply $0.01 < P(T > 2.30) < 0.025$. Symmetry gives $0.01 < P(T < -2.30) < 0.025$. Taking the complement gives $0.975 < P(T > -2.30) < 0.99$.

2.9 Interpret verbal probabilities for small business costs.

A standardized mean operating-cost model uses 6 businesses.

Questions/Tasks

C8 Find the positive cutoff exceeded by the standardized sample mean in 5 percent of model outcomes. Then find the symmetric cutoffs with 5 percent of outcomes outside them in total. Finally, find the symmetric cutoffs enclosing the middle 80 percent of outcomes.

C8 Solution

$T \sim t_\nu$, where $\nu = n - 1 = 6 - 1 = 5$. (a) One right tail: $\alpha = 0.05$; $t^*_{0.05,5} = 2.015$, giving a positive cutoff of approximately 2.015. (b) Two tails: $\alpha = 0.05/2 = 0.025$; $t^*_{0.025,5} = 2.571$, so the cutoffs are approximately ± 2.571. (c) Central area: $\alpha = (1 - 0.80)/2 = 0.10$; $t^*_{0.10,5} = 1.476$, so the cutoffs are approximately ± 1.476.

2.10 Determine the sample size for payroll records.

A wage study selects 5 employees from each of 5 departments. Treat all 25 observations as independent; the standardized mean follows the sample-mean model described above.

Questions/Tasks

C8 Find the negative cutoff below which the standardized sample mean falls in 1 percent of model outcomes. Then find the symmetric cutoffs enclosing the middle 90 percent of outcomes.

C8 Solution

$T \sim t_\nu$, where $\nu = n - 1 = 25 - 1 = 24$. $n = 5 \times 5 = 25$, so $\nu = 24$. For the left tail, use $\alpha = 0.01$ and symmetry: $t^*_{0.01,24} = 2.492$, so $c \approx -2.492$. For the central area, $\alpha = (1 - 0.90)/2 = 0.05$, giving $t^*_{0.05,24} = 1.711$ and approximate cutoffs ± 1.711.

2.11 Find probability ranges for revenue planning.

A standardized mean-revenue model uses samples of 10 stores.

Questions/Tasks



C8 For symmetric cutoffs of negative 3.00 and 3.00, give lower and upper bounds for the probability that the standardized sample mean falls below negative 3.00, the combined probability that it falls outside the cutoffs, and the probability that it lies between them. Also use the table to estimate the probability that the standardized sample mean lies between negative 1.833 and 2.262.

C8 Solution

$T \sim t_\nu$, where $\nu = n - 1 = 10 - 1 = 9$. Since $2.821 < 3.00 < 3.250$, $0.005 < P(T > 3.00) < 0.01$. Thus (a) $0.005 < P(T < -3.00) < 0.01$; (b) $0.01 < P(|T| > 3.00) < 0.02$; and (c) $0.98 < P(-3.00 < T < 3.00) < 0.99$. For the asymmetric interval, the left excluded tail is approximately 0.05 and the right excluded tail is approximately 0.025. Therefore $P(-1.833 < T < 2.262) \approx 1 - 0.05 - 0.025 = 0.925$.

